MILITARY SPECIFICATION SHEET

NUTS, SHEAR, SELF-LOCKING, STEEL, 180 KSI Ftu, 450°F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the bolts described herein shall consist of this document and the issue in effect of MIL-B-8985.

DESIGN INFORMATION:

The nuts covered by this specification sheet are for use where shear loads (not tensile) are the primary consideration.

The nuts are designed to be used with bolts conforming to MIL-B-8906/2, and with washers conforming to MS21299.

The nuts are subject to the limitations specified in MS33588.

Tooling shall be in accordance with MS33787 and MIL-W-8982.

REQUIREMENTS:

MATERIAL: ALLOY STEEL, AMS6322D, AMS6304, AMS6485, AMS6487 AND AMS6414.

HARDNESS: ROCKWELL C38-C46

SURFACE TEXTURE: BEARING SURFACE 125 MICRO INCHES IN ACCORDANCE WITH ANSI B46.1-1962.

PLATING: CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2.

AXIAL STRENGTH: SEATING TORQUE AND WRENCH TORQUE SHALL BE IN ACCORDANCE WITH TABLE I .

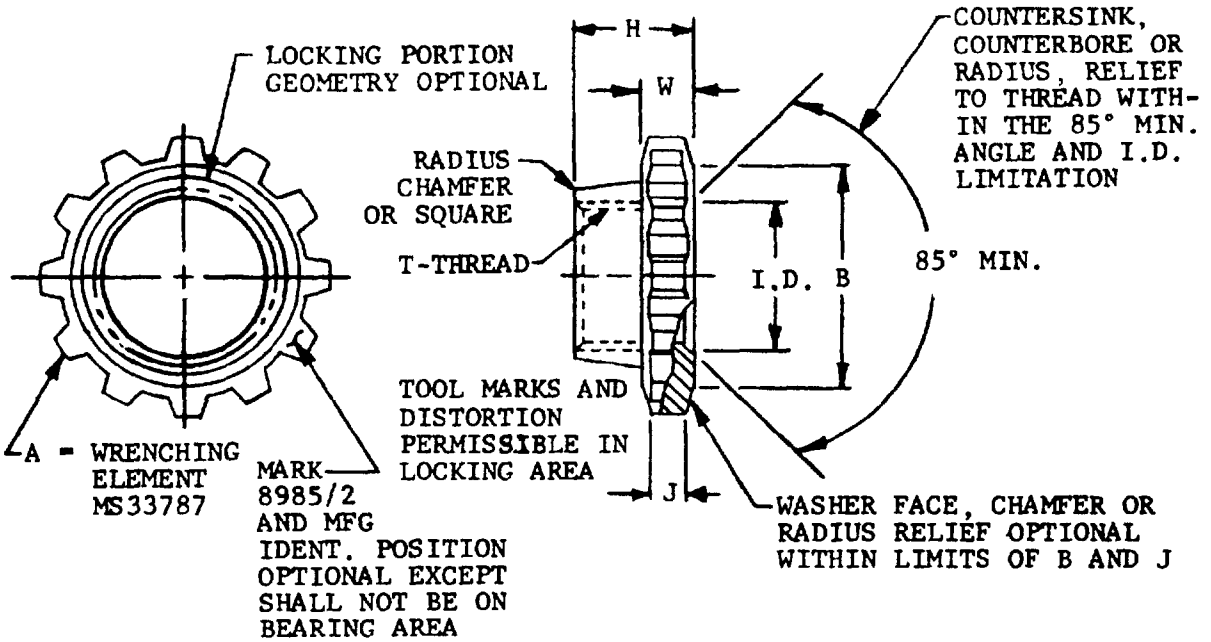
LUBRICATION: LUBRICATION SHALL BE IN ACCORDANCE WITH MIL-N-8985.

(SEE ILLUSTRATION ON PAGE 2)

FSC 5310

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:52Z
Check the source to verify that this is the current version before use.

ILLUSTRATION



AXIAL STRENGTH, SEATING TORQUE AND WRENCH TORQUF SHALL BE IN ACCORDANCE WITH TABLE II.

TABLE I

DASH NO.	ULTIMATF AXIAL STRENGTH LB. MIN. 2/ 3/	SEATING TORQUE FOR TORQUE EFFECTIVITY TEST IN LB	WRENCH TORQUF IN. LB. MIN.	MINIMUM BEARING AREA IN SQ INCHES 1/
-3	3,910	63	80	.0578
-4	6,980	150	200	.0735
-5	11,100	330	425	.0899
-6	17,100	540	700	0985
-7	23,200	790	1000	.1823
-8	30,900	1,100	1400	.1979
-9	39,200	1,530	1900	.2942
-10	49,000	2,000	2500	3343
-12	71,000	3,400	4400	.3737
-14	97,100	5,600	7100	.5435
-16	126,000	7,700	9800	.7444
-18	162,000	10,000	12500	.9737
-20	202,000	12,200	15500	1 1857
-22	247,000	14,400	17000	1 4985
-24	296,000	16,200	20500	1.8438

1/ BEARING AREA DATA ARE FOR DESIGN PURPOSES ONLY AND ARE NOT MANUFACTURING REQUIREMENTS

2/ SHALL BE TESTED ON 220 KSI BOLTS

3/ STRENGTH BASED ON NAS1348 SECTIONAL AREAS

Source: https://assist.dla.mil -- Downloaded: 202610-17T11:52Z
Check the source to verify that this is the current version before use.

TABLE II 1/

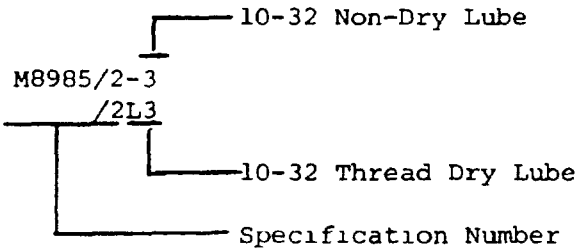
DASH NUMBER		THREAD UNJF-3B	MS33787 ELEMENT NUMBER	B MIN	I D		H MAX.	J MIN.	W MIN.	X SEE NOTE 2/	APPROX. WEIGHT LB/100
DRY LUBE	NON DRY LUBE				MAX.	MIN.					
L3	-3	10-32	12	343	.210	.190	.151	.024	.060	.003	.24
L4	-4	1/4-28	14	.408	.270	.250	.199	.049	.088	.003	42
L5	-5	5/16-24	16	474	332	.312	.245	.072	.113	.004	.65
L6	-6	3/8-24	18	.538	.405	375	.294	.096	.140	.004	99
L7	-7	7/16-20	22	.671	.467	.437	344	.110	.157	.005	1 75
L8	-8	1/2-20	24	.730	.530	.500	.394	.128	.181	.005	2.32
L9	-9	9/16-18	28	.855	.597	.562	447	.136	.196	.005	3.53
L10	-10	5/8-18	30	.928	.660	.625	.499	.155	.216	.006	4.41
L12	-12	3/4-16	34	1.042	.781	.750	.608	.301	.371	.007	8.12
L14	-14	7/8-14	40	1.230	.906	.875	.717	.331	.417	.008	12.67
L16	-16	1-12	46	1.418	1.031	1.000	.829	.387	.463	.009	19.20
L18	-18	1 1/8-12	52	1.605	1.156	1.125	945	.457	.568	.010	28.88
L20	-20	1 1/4-12	56	1.775	1.281	1 250	1.060	539	633	.011	36.81
L22	-22	1 3/8-12	62	1.971	1.406	1.375	1.187	.656	.760	.012	53.30
L24	-24	1 1/2-12	68	2.166	1.531	1.500	1 311	.744	.884	.013	69.30

NOTES 1/ Dimensions in inches. All dimensions after plating.
2/ Bearing surface shall be normal with pitch diameter of thread with X checked in accordance with MIL-N-8985.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:52Z
Check the source to verify that this is the current version before use.

PART NUMBER: Consists of the basic number of this specification sheet and a dash number taken from table I. The letter "L" in place of dash number indicates a dry lubricant has been added to the nut.

EXAMPLE:



CUSTODIANS:

ARMY - AV
AIR FORCE - 11

REVIEW

AIR FORCE - 82
DSA - IS

PREPARING ACTIVITY

NAVY - AS

DOD PROJECT NO. 5310-0897

Source: https://assist.dla.mil -- Downloaded: 202610-17T11:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

MIL-N-8985/2

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:52Z
Check the source to verify that this is the current version before use.